

Foundations and History of Artificial Intelligence

Foundations of AI

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Adapted from Russel & Norvig, ch. 1.2

1. Philosophy

- ➡ How can reasoning be formalized?
- ➡ How does the mind arise?
- ➡ Where does knowledge come from?
- ➡ How does knowledge lead to action?
- Interesting quote: “(the Pascaline calculator) produces effects which appear nearer to thought than all the actions of animals” Blaise Pascal (1642)

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2. Mathematics

- ➡ What are the formal rules to draw valid conclusions? (formal logic)
- ➡ What can be computed?
- ➡ How do we reason with uncertain information?

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3. Economics

- ➡ How should we make decisions in accordance to our preferences?
- ➡ How should we do this under adversary interests?
- ➡ How should we account for delayed payoffs?

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4. Neuroscience

➡ How do brains process information?

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5. Psychology

➡ How do humans and animals think and act?

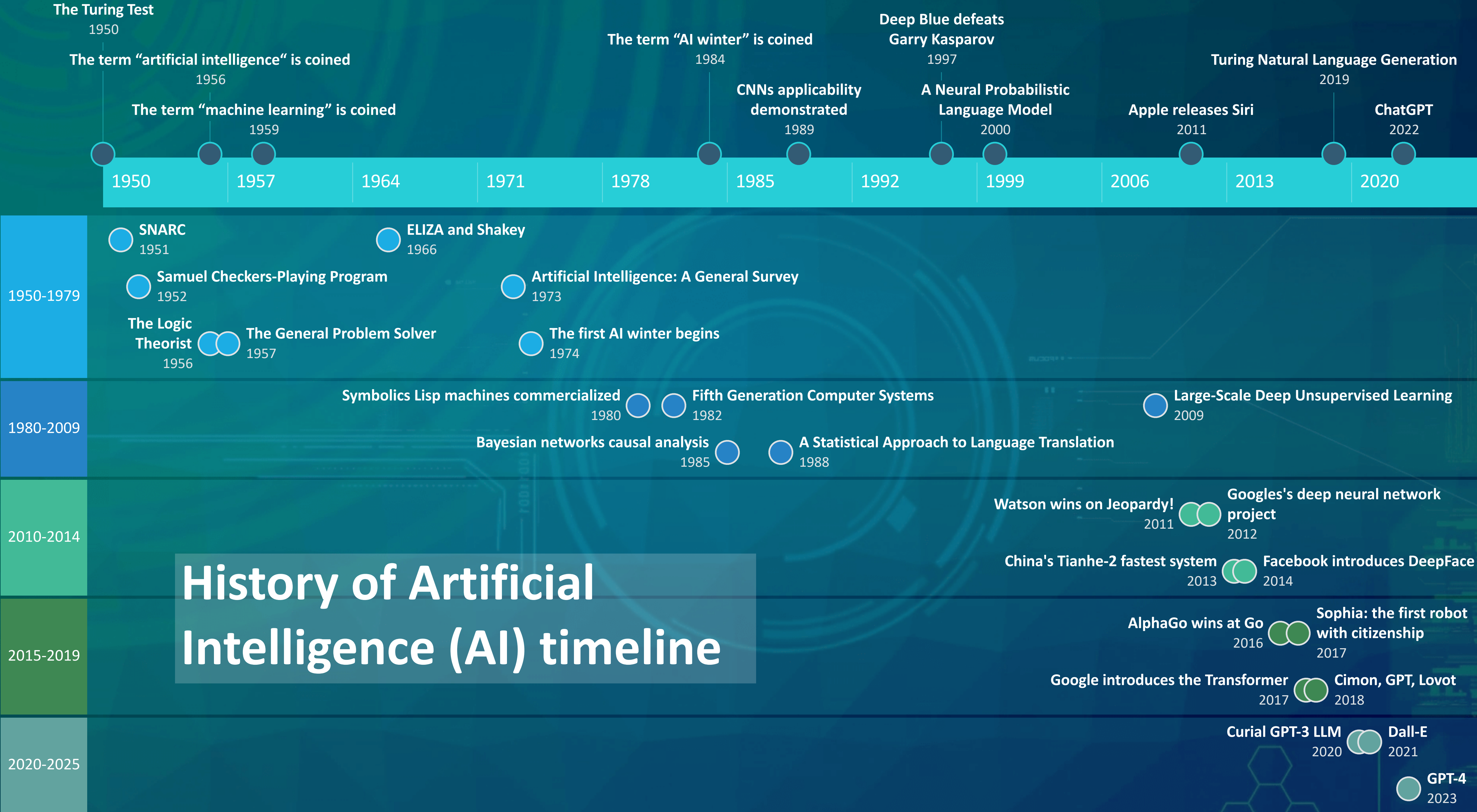
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6. Control Theory

➡ How can artifacts operate under their own control?

History of AI



Before the term “Artificial Intelligence”

Early works now recognized as “Artificial Intelligence”

- Artificial Neurons (McChulloch and Pitts, 1943)
- “Hebbian” learning in neurons (Hebb, 1949)
- Chess programs (Shannon 1950, Turing 1953)
 - ➡ Before there was a computer to run! Also mentioned in the “Turing Test” paper
- Neural Computer (Minsky and Edmonds, 1951)

The “birth” of AI

1956 Dartmouth Workshop



The “birth” of AI

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We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of **the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.** An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. **We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer**

The “Look, Ma, no hands!” era

Fast progress in the late 50s, early 60s

“A machine can never do X” - great progress in several X’es:

- General Problem Solver (Newell and Simon)
- Geometry Theorem Prover (Gelernter 1959)
- LISP language (McCarthy, 1958), “Programs with Common Sense” (1960)
- Microworlds, Blocks world (Minsky and several students)
- Perceptron, Perceptron Convergence Theorem (Block et al, 1962)

The first “AI Winter”

A dose of reality 1966-1973

Failure to scale to larger problems

- Most systems were based on “informed introspection” on how humans perform a task
- Lack of appreciation for intractability (“combinatorial explosion”) of many problems
 - No modern theory of computational complexity yet (NP-completeness by Cook, Levin 1972)
- Lack of representation power of perceptrons - can’t express a XOR function (Minsky, Papert 1969)
- Bold predictions: A computer will be chess champion in 10 years - it took 40

The first “AI Winter”

A dose of reality 1966-1973

Consequences

- Lighthill Report (Lighthill 1973) to the UK Parliament found AI had utterly failed to achieve its “grandiose objectives”
- DARPA funding cuts after five-year experiment in speech understanding failed to meet expectations.

Expert Systems

Incorporation of domain knowledge 1969-1986

- DENDRAL (Buchanan 1969) used known patterns to derive molecular structure from the elementary formula of a molecule
- MYCIN diagnosed blood infections with 450 rules
- R1, first commercially successful expert system reportedly saved DEC \$40 million a year helping configure orders for new computer systems
- Wide proliferation in industry: millions of dollars in 1980 -> billions in 1988

Focus shift from general, foundational problem-solving to specific, economically valuable problems

The Second AI Winter

Expert knowledge meets bottleneck late 80s, early 90s

- Expert systems found brittle, hard to scale, maintain and update.
- Collapse of the LISP machine market in favor of new, more general architectures
- Failure of the Japanese “Fifth Generation Computer” (1981-1991) leading to more DARPA funding cuts

Probabilistic reasoning and return of neural networks

1986 - Present

Focus on approaches based on rigorous mathematical, validation with shared benchmark problem sets (LibriSpeech, MNIST, ImageNet, COCO....)

- Back-propagation “reinvented” by four different groups (mid-80s)
- Hidden Markov Models outperform ad-hoc methods in speech recognition (mid-80s)
- CNN Applicability to computer vision
- Universal Approximation Theorems (e.g. Cybenko 1989) for neural networks with non-linear activation functions
 - Solves the perceptron problem!

Computers outperform humans in Chess

Deep Blue (IBM) vs Gary Kasparov (1996)



The deep Learning Era

2000s - present

- Formalization Reinforcement Learning with Markov Decision processes (Sutton 1998)
- Big Data” leverages availability of data in the WEB
 - IBM Watson beats human Jeopardy! champions (IBM, 2010)
- Deep Learning dominates computer vision and speech (2011 - present)
 - Neural networks with many neurons, organized in several layers
- Alpha Go beats Lee Sedol in Go (2016) using reinforcement learning and deep NNs
- “Attention is all you need” (Google, 2017) introduces the Transformer

The deep Learning Era (cont.)

The LLM race

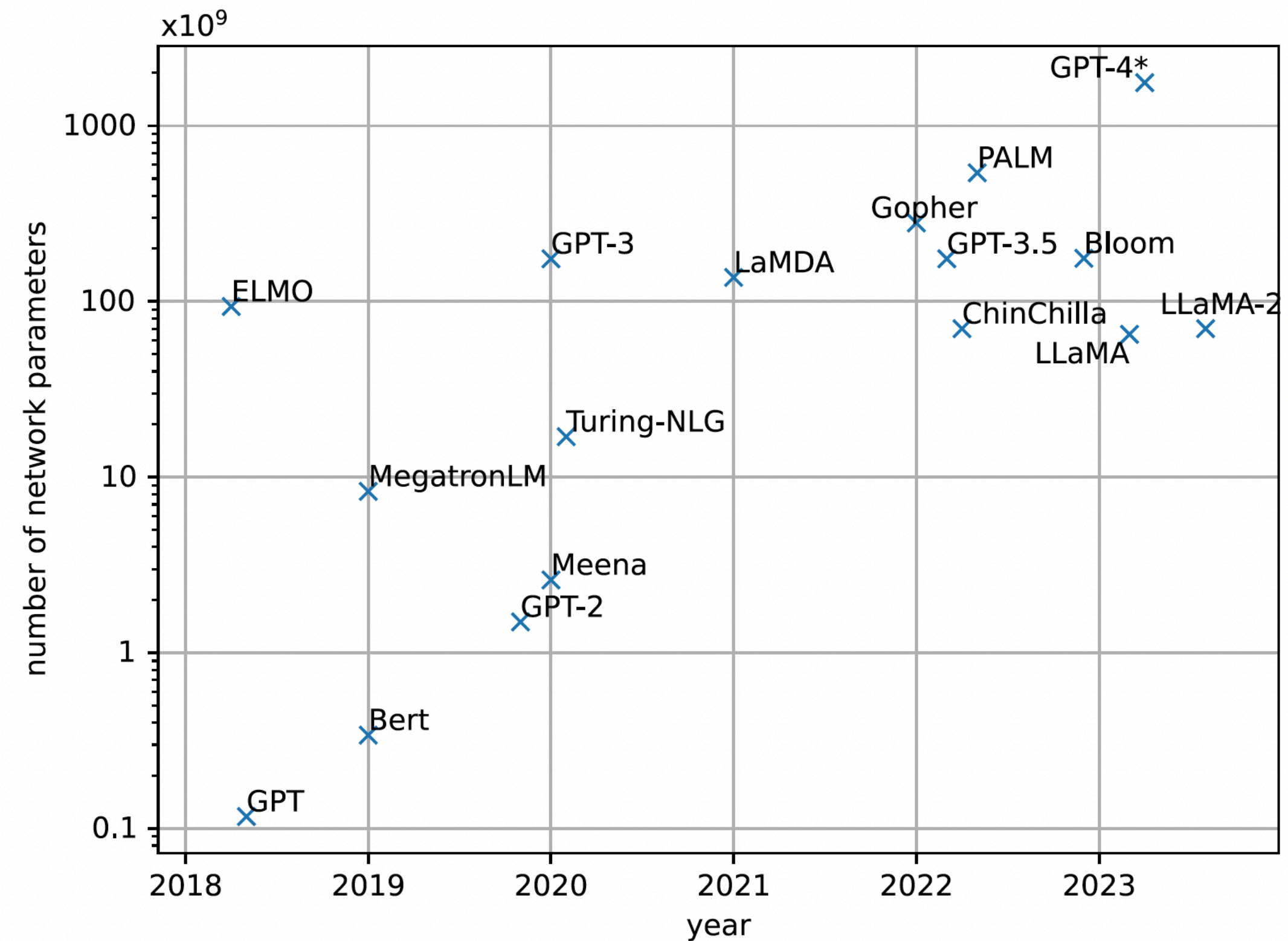


Fig. 1 Number of parameters of LLM over the past five years. Significant advances were made by Megatron in 2019 and GPT-3 in 2020. *The number of parameters of GPT-4 is not officially disclosed but was presumably leaked

Source: Multibody Models Generated from Natural Language. Gerstmayr et. al 2023

url: <https://link.springer.com/article/10.1007/s11044-023-09962-0> (access 01/09/2926)